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# Brazilian Company Data Analysis

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## Tech Stack

**Python | Numpy | Pandas | Matplotlib | Seaborn | MySQL**

## List all Unique cities where Customers are Located

[2]:
```
query = """ select distinct customer_city from customers """
cur.execute(query)
data = cur.fetchall()
print(len(data))
data
```

```
4119
```

[2]:
```
[('franca',),
 ('sao bernardo do campo',),
 ('sao paulo',),
 ('mogi das cruzes',),
 ('campinas',),
 ('jaragua do sul',),
 ('timoteo',),
 ('curitiba',),
 ('belo horizonte',),
 ('montes claros',),
 ('rio de janeiro',),
 ('lencois paulista',),
 ('caxias do sul',),
 ('piracicaba',),
 ('guarulhos',),
 ('pacaja',),
```

## Count the Number of Orders placed in 2017

[3]:
```
query = """ select count(order_id) from orders where year(order_purchase_timestamp) = 2017"""
cur.execute(query)
data=cur.fetchall()
"Total orders placed in 2017 are: ",data[0][0]
```

[3]:
```
('Total orders placed in 2017 are: ', 45101)
```

[4]:
```
query= """ select count(order_id) from orders where year(order_purchase_timestamp)=2018"""
cur.execute(query)
data=cur.fetchall()
"Total orders placed in 2018 are: ",data[0][0]
```

[4]:
```
('Total orders placed in 2018 are: ', 54011)
```

[5]:
```
query = """SELECT DISTINCT YEAR(order_purchase_timestamp) FROM orders"""
cur.execute(query)
data = cur.fetchall()

print(len(data))
data
```

```
3
```

[5]:
```
[(2017,), (2018,), (2016,)]
```

## Find the Total Sales per Category ¶

[7]:
```
query = """ select upper(products.product_category) category,
round(sum(payments.payment_value),2) sales
from products join order_items
on products.product_id = order_items.product_id
join payments
on payments.order_id = order_items.order_id
group by category
"""

cur.execute(query)

data = cur.fetchall()

df = pd.DataFrame(data, columns = ["Category", "Sales"])
df
```

[7]:

| | Category | Sales |
|---|---|---|
| **0** | PERFUMERY | 506738.66 |
| **1** | FURNITURE DECORATION | 1430176.39 |
| **2** | TELEPHONY | 486882.05 |
| **3** | BED TABLE BATH | 1712553.67 |
| **4** | AUTOMOTIVE | 852294.33 |
| **...** | ... | ... |
| **69** | CDS MUSIC DVDS | 1199.43 |
| **70** | LA CUISINE | 2913.53 |
| **71** | FASHION CHILDREN'S CLOTHING | 785.67 |
| **72** | PC GAMER | 2174.43 |
| **73** | INSURANCE AND SERVICES | 324.51 |

74 rows × 2 columns

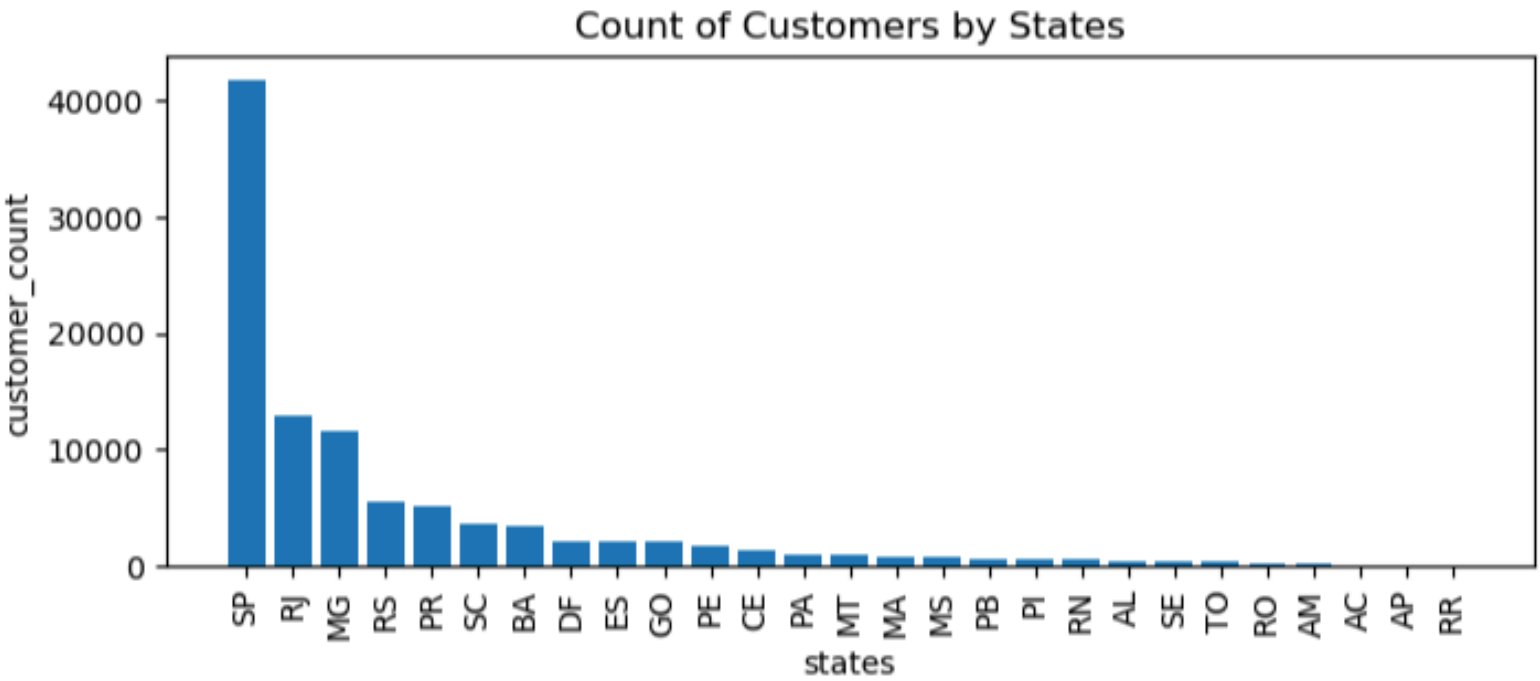
## Percentage of Orders Paid in Installments. ¶

[8]:
```
query = """ select ((sum(case when payment_installments >= 1 then 1
else 0 end))/count(*))*100 from payments
"""

cur.execute(query)

data = cur.fetchall()

"the percentage of orders that were paid in installments is", data[0][0]
```

[8]:
```
('the percentage of orders that were paid in installments is',
 Decimal('99.9981'))
```

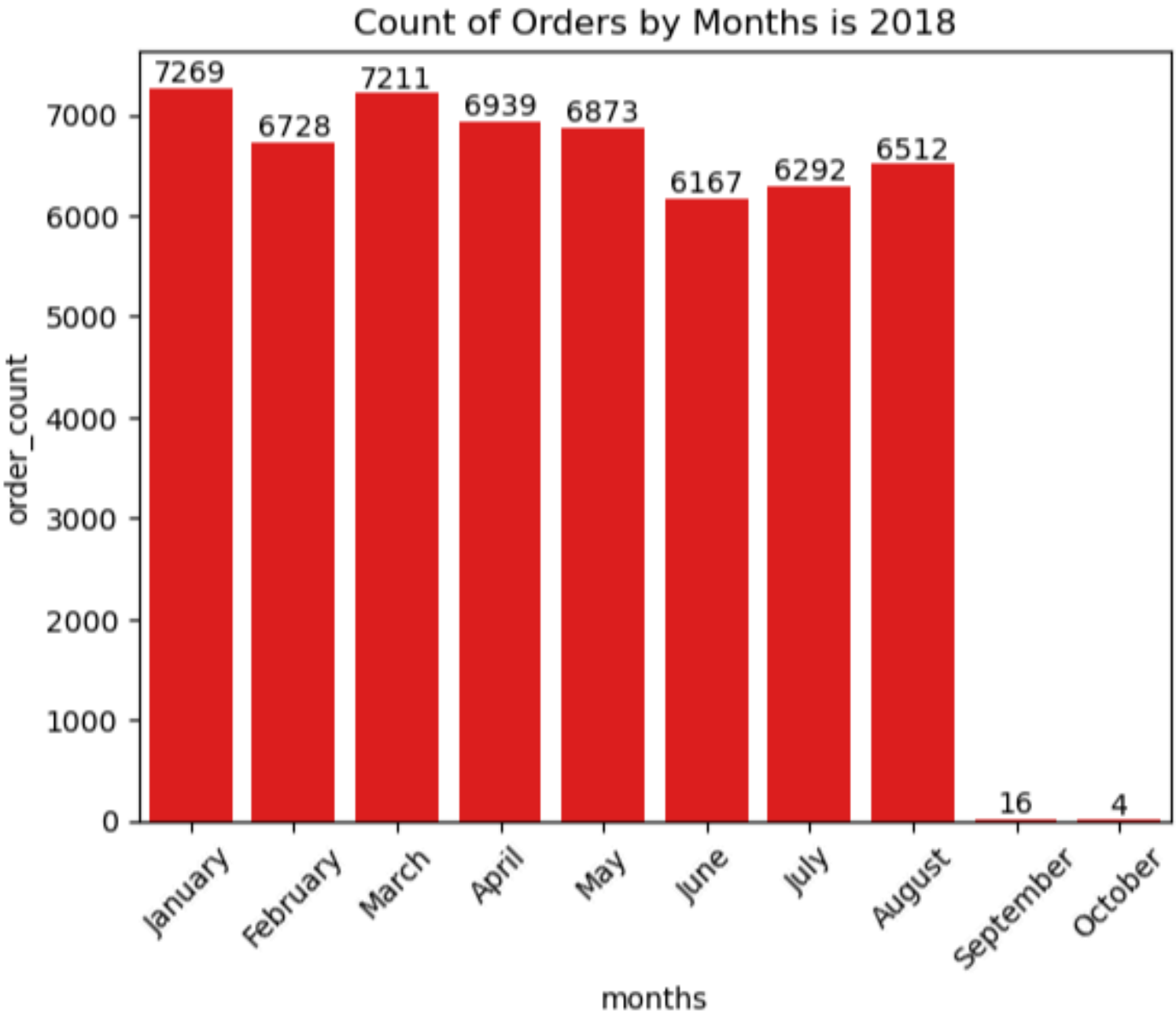
## Number of Customers from each State.

[9]:
```
query = """ select customer_state ,count(customer_id)
from customers group by customer_state
"""

cur.execute(query)

data = cur.fetchall()
df = pd.DataFrame(data, columns = ["state", "customer_count" ])
df = df.sort_values(by = "customer_count", ascending= False)

plt.figure(figsize = (8,3))
plt.bar(df["state"], df["customer_count"])
plt.xticks(rotation = 90)
plt.xlabel("states")
plt.ylabel("customer_count")
plt.title("Count of Customers by States")
plt.show()
```



## Number of Orders per Month in 2018.

[10]:
```
query = """ select monthname(order_purchase_timestamp) months, count(order_id) order_count
from orders where year(order_purchase_timestamp) = 2018
group by months
"""

cur.execute(query)

data = cur.fetchall()
df = pd.DataFrame(data, columns = ["months", "order_count"])
o = ["January", "February","March","April","May","June","July","August","September","October"]

ax = sns.barplot(x = df["months"],y =  df["order_count"], data = df, order = o, color = "red")
plt.xticks(rotation = 45)
ax.bar_label(ax.containers[0])
plt.title("Count of Orders by Months is 2018")

plt.show()
```



## Average Number of Products Per Order, Grouped by Customer City.

[11]:
```
query = """with count_per_order as
(select orders.order_id, orders.customer_id, count(order_items.order_id) as oc
from orders join order_items
on orders.order_id = order_items.order_id
group by orders.order_id, orders.customer_id)

select customers.customer_city, round(avg(count_per_order.oc),2) average_orders
from customers join count_per_order
on customers.customer_id = count_per_order.customer_id
group by customers.customer_city order by average_orders desc
"""

cur.execute(query)

data = cur.fetchall()
df = pd.DataFrame(data,columns = ["customer city", "average products/order"])
df.head(10)
```

| | customer city | average products/order |
|---|---|---|
| 0 | padre carvalho | 7.00 |
| 1 | celso ramos | 6.50 |
| 2 | candido godoi | 6.00 |
| 3 | datas | 6.00 |
| 4 | matias olimpio | 5.00 |
| 5 | morro de sao paulo | 4.00 |
| 6 | cidelandia | 4.00 |
| 7 | picarra | 4.00 |
| 8 | teixeira soares | 4.00 |
| 9 | curralinho | 4.00 |

## Percentage of Total Revenue Contributed by each Product Category.

```
query = """select upper(products.product_category) category,
round((sum(payments.payment_value)/(select sum(payment_value) from payments))*100,2) sales_percentage
from products join order_items
on products.product_id = order_items.product_id
join payments
on payments.order_id = order_items.order_id
group by category order by sales_percentage desc"""


cur.execute(query)
data = cur.fetchall()
df = pd.DataFrame(data,columns = ["Category", "percentage distribution"])
df.head()
```

| | Category | percentage distribution |
|---|---|---|
| 0 | BED TABLE BATH | 10.70 |
| 1 | HEALTH BEAUTY | 10.35 |
| 2 | COMPUTER ACCESSORIES | 9.90 |
| 3 | FURNITURE DECORATION | 8.93 |
| 4 | WATCHES PRESENT | 8.93 |

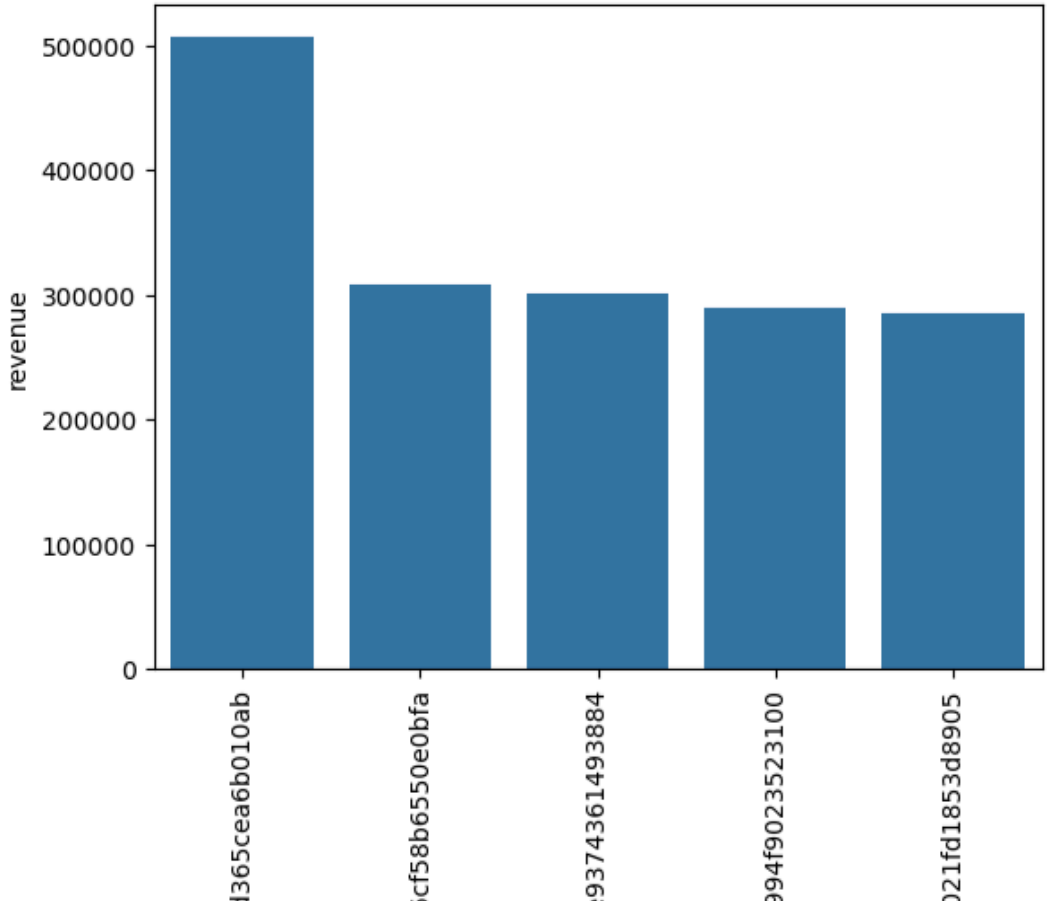
## Correlation between Product price and the Number of Times a Product has been Purchased.

```
cur = db.cursor()
query = """select products.product_category,
count(order_items.product_id),
round(avg(order_items.price),2)
from products join order_items
on products.product_id = order_items.product_id
group by products.product_category"""

cur.execute(query)
data = cur.fetchall()
df = pd.DataFrame(data,columns = ["Category", "order_count","price"])

arr1 = df["order_count"]
arr2 = df["price"]

a = np.corrcoef([arr1,arr2])
print("the correlation is", a[0][-1])
```

```
the correlation is -0.10631514167157562
```

## Total Revenue Generated by each Seller, and Rank them by Revenue.

```
query = """ select *, dense_rank() over(order by revenue desc) as rn from
(select order_items.seller_id, sum(payments.payment_value)
revenue from order_items join payments
on order_items.order_id = payments.order_id
group by order_items.seller_id) as a """

cur.execute(query)
data = cur.fetchall()
df = pd.DataFrame(data, columns = ["seller_id", "revenue", "rank"])
df = df.head()
sns.barplot(x = "seller_id", y = "revenue", data = df)
plt.xticks(rotation = 90)
plt.show()
```



## The Moving Average of Order Values for each Customer over their order History. ¶

```
query = """select customer_id, order_purchase_timestamp, payment,
avg(payment) over(partition by customer_id order by order_purchase_timestamp
rows between 2 preceding and current row) as mov_avg
from
(select orders.customer_id, orders.order_purchase_timestamp,
payments.payment_value as payment
from payments join orders
on payments.order_id = orders.order_id) as a"""
cur.execute(query)
data = cur.fetchall()
df = pd.DataFrame(data)
df
```

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| **0** | 00012a2ce6f8dcda20d059ce98491703 | 2017-11-14 16:08:26 | 114.74 | 114.739998 |
| **1** | 000161a058600d5901f007fab4c27140 | 2017-07-16 09:40:32 | 67.41 | 67.410004 |
| **2** | 0001fd6190edaaf884bcaf3d49edf079 | 2017-02-28 11:06:43 | 195.42 | 195.419998 |
| **3** | 0002414f95344307404f0ace7a26f1d5 | 2017-08-16 13:09:20 | 179.35 | 179.350006 |
| **4** | 000379cdec625522490c315e70c7a9fb | 2018-04-02 13:42:17 | 107.01 | 107.010002 |
| **...** | ... | ... | ... | ... |
| **103881** | fffecc9f79fd8c764f843e9951b11341 | 2018-03-29 16:59:26 | 71.23 | 27.120001 |
| **103882** | fffeda5b6d849fbd39689bb92087f431 | 2018-05-22 13:36:02 | 63.13 | 63.130001 |
| **103883** | ffff42319e9b2d713724ae527742af25 | 2018-06-13 16:57:05 | 214.13 | 214.130005 |
| **103884** | ffffa3172527f765de70084a7e53aae8 | 2017-09-02 11:53:32 | 45.50 | 45.500000 |
| **103885** | ffffe8b65bbe3087b653a978c870db99 | 2017-09-29 14:07:03 | 18.37 | 18.370001 |

103886 rows × 4 columns

## Cumulative Sales per Month for each Year.

```
query = """select years, months , payment, sum(payment)
over(order by years, months) cumulative_sales from
(select year(orders.order_purchase_timestamp) as years,
month(orders.order_purchase_timestamp) as months,
round(sum(payments.payment_value),2) as payment from orders join payments
on orders.order_id = payments.order_id
group by years, months order by years, months) as a
"""
cur.execute(query)
data = cur.fetchall()
df = pd.DataFrame(data)
df
```

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| **0** | 2016 | 9 | 252.24 | 252.24 |
| **1** | 2016 | 10 | 59090.48 | 59342.72 |
| **2** | 2016 | 12 | 19.62 | 59362.34 |
| **3** | 2017 | 1 | 138488.04 | 197850.38 |
| **4** | 2017 | 2 | 291908.01 | 489758.39 |
| **5** | 2017 | 3 | 449863.60 | 939621.99 |

## Year-over-Year Growth rate of Total Sales.

```
query = """with a as(select year(orders.order_purchase_timestamp) as years,
round(sum(payments.payment_value),2) as payment from orders join payments
on orders.order_id = payments.order_id
group by years order by years)

select years, ((payment - lag(payment, 1) over(order by years))/
lag(payment, 1) over(order by years)) * 100 from a"""

cur.execute(query)
data = cur.fetchall()
df = pd.DataFrame(data, columns = ["years", "yoy % growth"])
df
```

| | years | yoy % growth |
|---|---|---|
| **0** | 2016 | NaN |
| **1** | 2017 | 12112.703761 |
| **2** | 2018 | 20.000924 |

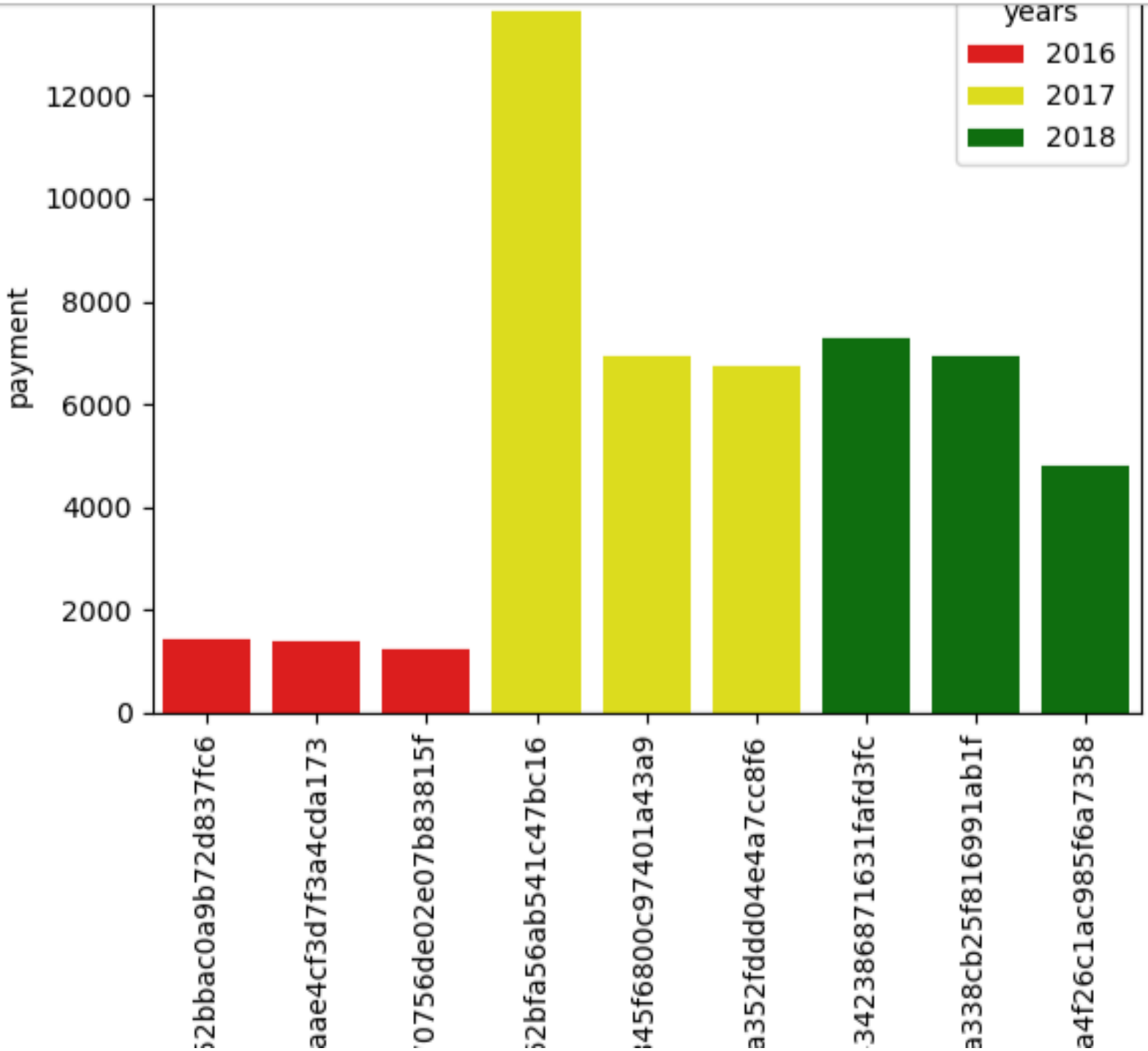
## Retention rate of Customers, defined as the Percentage of Customers who make another Purchase within 6 Months of their First Purchase.

```
query = """with a as (select customers.customer_id,
min(orders.order_purchase_timestamp) first_order
from customers join orders
on customers.customer_id = orders.customer_id
group by customers.customer_id),

b as (select a.customer_id, count(distinct orders.order_purchase_timestamp) next_order
from a join orders
on orders.customer_id = a.customer_id
and orders.order_purchase_timestamp > first_order
and orders.order_purchase_timestamp <
date_add(first_order, interval 6 month)
group by a.customer_id)

select 100 * (count( distinct a.customer_id)/ count(distinct b.customer_id))
from a left join b
on a.customer_id = b.customer_id ;"""

cur.execute(query)
data = cur.fetchall()

data
```

```
[(None,)]
```

## Top 3 Customers who Spent the Most Money in each Year.

```
query = """select years, customer_id, payment, d_rank
from
(select year(orders.order_purchase_timestamp) years,
orders.customer_id,
sum(payments.payment_value) payment,
dense_rank() over(partition by year(orders.order_purchase_timestamp)
order by sum(payments.payment_value) desc) d_rank
from orders join payments
on payments.order_id = orders.order_id
group by year(orders.order_purchase_timestamp),
orders.customer_id) as a
where d_rank <= 3 ;"""

cur.execute(query)
data = cur.fetchall()

df = pd.DataFrame(data, columns=["years", "id", "payment", "rank"])

# Define custom colors (adjust depending on number of years)
custom_palette = ["red", "yellow", "green"]

sns.barplot(x="id", y="payment", data=df, hue="years", palette=custom_palette)

plt.xticks(rotation=90)
plt.show()
```


years
2016
2017
2018
payment
0
2000
4000
6000
8000
10000
12000
52bbac0a9b72d837fc6
aae4cf3d7f3a4cda173
0756de02e07b83815f
52bfa56ab541c47bc16
45f6800c97401a43a9
a352fddd04e4a7cc8f6
342386871631fafd3fc
a338cb25f816991ab1f
a4f26c1ac985f6a7358